

A Noise-Aware Cost Function for Multi-Qubit Gate Decomposition Selection on IBM Heron Devices

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Abstract

Many multi-qubit gates (CCX, CCZ, SWAP, ...) admit several algebraically equivalent decompositions into a device's native basis, with different gate counts, depths, and error footprints. This note formulates a cost function that scores each candidate decomposition using backend calibration data, so that the transpiler can select the decomposition with the lowest predicted error for the *specific physical qubits* it will run on. We define the cost function in two versions: v0, an additive per-gate error model, and v1, which augments v0 with idle-time decoherence penalties derived from the thermal relaxation channel. Every input variable is extractable from **BackendV2** calibration data for the Heron processor family. We state all assumptions explicitly, fix policies for calibration edge cases, and propose a validation protocol against Qiskit Aer noise-model simulation.

1 Setting and problem statement

Fix a Heron-family backend with native gate set $\{ \text{CZ}, \text{RZ}(\theta), \text{SX}, \text{X}, \text{ID} \}$ (plus fractional RX, RZZ where enabled). For a target gate G (e.g. Toffoli), let $\mathcal{D}(G) = \{D_1, \dots, D_m\}$ be a catalogue of circuits over the native basis, each implementing G exactly (or up to a stated relative phase). Given a placement of G 's logical qubits onto physical qubits, each D_k becomes a scheduled circuit on specific physical qubits and coupling-map edges.

Goal. Define a cost function $C(D)$, computable from backend calibration data alone, such that selecting $\arg \min_k C(D_k)$ maximises the expected fidelity of the implemented gate. The transpiler pass built by Group B will evaluate C over the catalogue at compile time.

2 Physical background: what determines gate fidelity

2.1 Average gate fidelity

An ideal gate is a unitary transformation U but the hardware implements a noisy quantum channel $\mathcal{E}$. The *average gate fidelity* compares the two, averaged uniformly (Haar measure) over pure input states:

$$\bar{F}(\mathcal{E}, U) = \int d\psi \langle \psi | U^\dagger \mathcal{E}(\psi) U | \psi \rangle. \quad (1)$$

Nielsen [1] gives the closed form $\bar{F} = (d F_e + 1)/(d + 1)$ in dimension d , where F_e is the entanglement fidelity of $\mathcal{E} \circ U^\dagger$. The reported *error rate* of a gate is $\epsilon = 1 - \bar{F}$.

The average-case definition (rather than a worst-case one) is the operationally relevant quantity because it is what *randomized benchmarking* (RB) estimates [2]: the exponential decay of survival probability over random Clifford sequences yields the average error per gate, with state-preparation and measurement errors absorbed into the fit constants. IBM's daily benchmarking publishes exactly these RB-derived error rates through the backend calibration data.

2.2 Error mechanisms

Three mechanisms dominate:

1. **Decoherence during the gate.** Every gate has finite duration t . Energy relaxation (amplitude damping, timescale T_1) and dephasing (phase damping, timescale $T_2 \leq 2T_1$) act throughout, giving a floor $\epsilon \gtrsim \mathcal{O}(t/T_1)$. This is the main reason two-qubit gates ($t \sim 10^2$ ns) are roughly an order of magnitude noisier than single-qubit gates, and why circuit *depth* matters independently of gate count.
2. **Control error.** Miscalibrated pulse amplitude or phase produces coherent over-/under-rotation. Calibration drifts on the timescale of hours (e.g. two-level-system defects coupling to individual qubits), which is why error rates are per-qubit, per-edge, and time-varying — and hence why noise-*adaptive* selection can beat any static decomposition rule.
3. **Crosstalk and leakage.** Simultaneous gates disturb neighbours, and transmons can leak to $|2\rangle$. These are *not* separately reported in calibration data; they are excluded from the model and listed as a limitation (§8). The gap between IBM’s isolated per-edge errors and the layered EPLG benchmark [5] quantifies their typical size.

3 Calibration data inventory (Heron)

All quantities below are extractable programmatically from BackendV2 (`backend.target` for per-instruction properties; `backend.qubit_properties` / `backend.properties()` for qubit physics). This inventory defines the interface contract with Group B’s extraction layer.

Symbol	Meaning	Granularity	Notes
$\epsilon_{\text{CZ}}(i, j)$	CZ error rate	per edge	RB in isolation; includes 1Q Clifford layer
$t_{\text{CZ}}(i, j)$	CZ duration	per edge	
$\epsilon_{1q}(i)$	SX/X error rate	per qubit	single RB value shared by SX, X, ID
$t_{1q}(i)$	SX/X duration	per qubit	
$T_1(i), T_2(i)$	coherence times	per qubit	T_2 from Hahn echo
$\epsilon_{\text{ro}}(i)$	readout error	per qubit	excluded from C ; see §6

Two structural facts shape the cost function:

- $\text{RZ}(\theta)$ is a *virtual* frame rotation performed in software; its reported error and duration are exactly zero. RZ gates therefore contribute nothing to cost and are dropped.
- Single-qubit physical gates on a given qubit share one RB error value, so the single-qubit contribution reduces to (number of SX/X gates) $\times \epsilon_{1q}$ per qubit.

4 Cost function v0: additive gate-error model

Assumption 1 (Independence). *Gate errors act independently across the gates of a decomposition i.e. the probability that the whole decomposition executes without error is the product of per-gate success probabilities.*

This is the standard working assumption of noise-adaptive compilation [3] and of layout-scoring tools such as *mapomatic* [4]; it neglects crosstalk (§8).

Under Assumption 1, a scheduled decomposition D succeeds with probability

$$F_{\text{gates}}(D) = \prod_{g \in D} (1 - \epsilon_g), \quad (2)$$

where ϵ_g is looked up for the specific physical qubit or edge on which g acts. Defining the cost as the negative log-fidelity makes it additive:

$$C_0(D) = - \sum_{g \in D} \ln(1 - \epsilon_g) \approx \sum_{g \in D} \epsilon_g \quad (\epsilon_g \ll 1). \quad (3)$$

Lower cost corresponds to higher predicted fidelity, and evaluating Eq. (3) over a catalogue is $\mathcal{O}(\text{gate count})$ per candidate. Because ϵ_{CZ} varies substantially edge to edge, C_0 is already noise-adaptive: two decompositions with identical gate counts generally receive different costs once placed on hardware.

5 Cost function v1: idle decoherence

C_0 is blind to depth: a decomposition with fewer gates but a longer critical path leaves qubits idle, and idle qubits decohere. We model idling as an identity gate implemented by the thermal relaxation channel and compute its average fidelity via Eq. (1).

5.1 Derivation of the idle penalty

For a qubit idle for time τ , amplitude damping decays the excited-state population as $e^{-\tau/T_1}$ and phase damping decays coherences as $e^{-\tau/T_2}$ (Markovian environment $\Rightarrow$ exponential laws; T_1, T_2 are defined as these $1/e$ times and measured accordingly). Haar-averaging the combined channel against the identity over the Bloch sphere gives the standard closed form

$$\bar{F}_{\text{idle}}(\tau) = \frac{1}{2} + \frac{1}{6} e^{-\tau/T_1} + \frac{1}{3} e^{-\tau/T_2}. \quad (4)$$

Sanity checks: $\bar{F}_{\text{idle}}(0) = 1$; $\bar{F}_{\text{idle}}(\infty) = \frac{1}{2}$, the average overlap of the relaxed state $|0\rangle$ with a Haar-random target. The coefficients reflect the Bloch-sphere geometry: dephasing attacks the two equatorial components, relaxation the polar one.

Idle windows ($\tau \sim \text{ns-}\mu\text{s}$) are short against Heron coherence times ($T_1, T_2 \sim 10^2 \mu\text{s}$), so expanding to first order and taking the negative log:

$$-\ln \bar{F}_{\text{idle}}(\tau) \approx \frac{\tau}{6 T_1} + \frac{\tau}{3 T_2}. \quad (5)$$

5.2 Definition of v1

Let τ_q denote the total idle time of physical qubit q in the scheduled decomposition (computed from the instruction durations in §3 and the circuit schedule). Then

$$C_1(D) = \underbrace{- \sum_{g \in D} \ln(1 - \epsilon_g)}_{\text{gate errors (v0)}} + \underbrace{\sum_q \left(\frac{\tau_q}{6 T_1(q)} + \frac{\tau_q}{3 T_2(q)} \right)}_{\text{idle decoherence}}. \quad (6)$$

The RB-derived ϵ_g already include decoherence *during* each gate; the second term adds only what happens *between* gates, so there is no double-counting at this order. One second-order caveat: relaxation itself contributes to dephasing ($T_2 \leq 2T_1$), so using the reported T_2 directly slightly double-weights the T_1 mechanism; this affects the two coefficients but not, generically, the induced ranking.

The idle term is what allows a shallower but slightly gate-heavier decomposition to beat a deep, skinny one — precisely the trade-off a noise-aware selector exists to adjudicate.

6 Edge-case policies

Deliberate, documented choices rather than accidents:

- **Stale calibrations.** If benchmarking of a qubit or edge fails over several days, IBM reports $\epsilon = 1$ (meaning “undefined”, not a literal error probability). Under Eq. (3) this gives $C = \infty$, i.e. the edge is never selected. We adopt this as intended behaviour and log a warning when it triggers.
- **Units.** The IBM Platform UI displays durations in nanoseconds; Qiskit returns *seconds*. All internal computation is in seconds.
- **Readout error.** Candidate decompositions of the same gate are measured identically downstream, so readout error is a constant offset that cannot change the ranking; it is excluded from C .
- **Relative-phase decompositions.** Catalogue entries correct only up to relative phase (e.g. Margolus) are scored identically but flagged, since their validity depends on downstream circuit context; the selection pass must respect the flag.
- **Fractional gates.** RZZ calibration is only exposed when the backend is loaded with `use_fractional_gates=True`; the extraction layer should load backends accordingly if fractional decompositions enter the catalogue.

7 Validation protocol

The cost function is a *ranking* device, so validation targets rank agreement, not absolute error prediction:

1. Build the CCX/Toffoli catalogue on a Heron fake backend (e.g. `FakeTorino`) across several physical-qubit placements.
2. Score every (decomposition, placement) pair with C_0 and C_1 .
3. Simulate each pair through the Aer noise model built from the same calibration snapshot; estimate output fidelity (state or process fidelity for the isolated gate; success probability for small benchmark circuits).
4. Report rank correlation (Spearman ρ) between predicted cost and observed error, for C_0 and C_1 separately. The marginal value of the idle term is measured by $\rho(C_1) - \rho(C_0)$, expected to be largest for placements/decompositions with strongly differing depths.

8 Limitations and v2 roadmap

- **Average-case ϵ .** RB errors are Haar-averaged (Eq. 1); a specific circuit feeds each gate a specific state, so average-case rates can mis-estimate the realised error. Shared with all calibration-data-driven compilation [3, 4].
- **Independence / crosstalk.** Assumption 1 ignores simultaneous-gate interference. The isolated-vs-layered (EPLG) gap [5] bounds the typical size of the neglected effect; a v2 correction could interpolate toward layered error rates.
- **Scalar error model.** ϵ compresses the Pauli error structure to one number; direction-dependence of two-qubit errors is likewise ignored. Both are candidate v2 refinements, contingent on data availability and check-in priorities.
- **Drift.** Costs are only as fresh as the calibration snapshot; ranking stability under drift is an open empirical question we can measure for free during validation.

References

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